

V. EDA – DATA VISUALIZATION

AIM:

To explore and understand the underlying patterns, distributions, and relationships within the dataset through visual representations, which aids in uncovering insights, detecting anomalies, and guiding further analysis.

```
[1]: import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
```

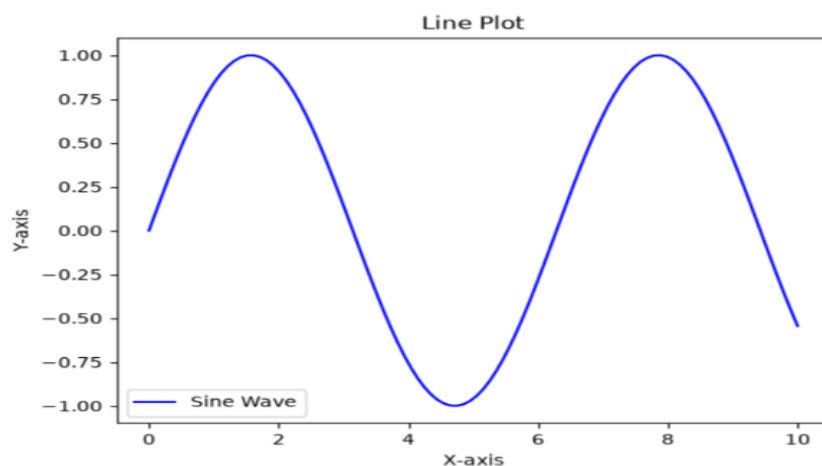
```
[2]: df = pd.read_csv("data.csv")
df.head()
```

```
[2]:
```

	Name	Age	Marks	Department
0	Arun	19	85	CSE
1	Priya	20	92	AIML
2	Kavin	19	78	ECE
3	Divya	21	88	CSE
4	Rahul	20	75	AIML

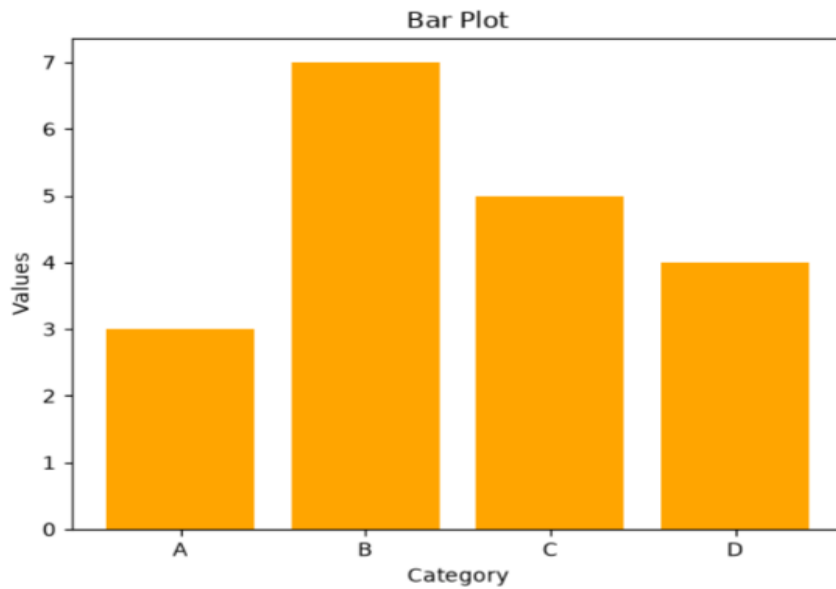
```
[3]: x = np.linspace(0, 10, 100)
y = np.sin(x)

plt.plot(x, y, color='blue', label='Sine Wave')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Line Plot')
plt.legend()
plt.show()
```



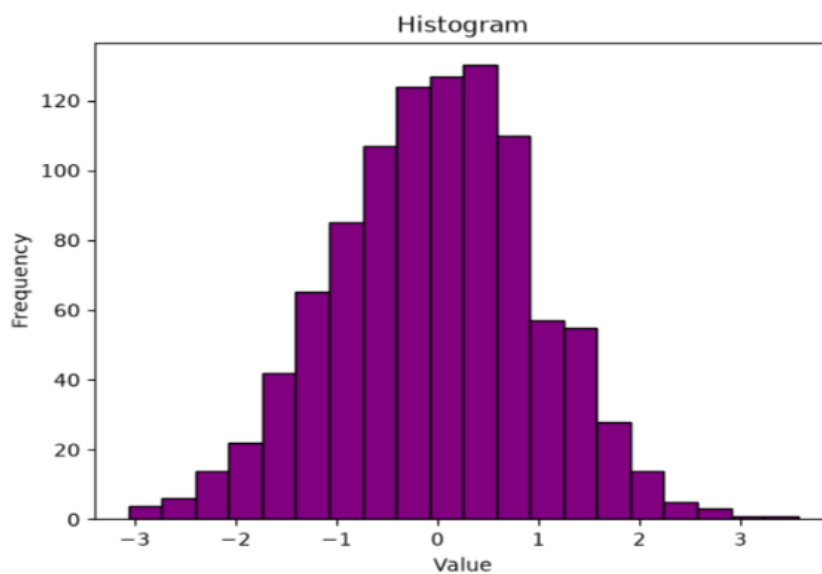
```
[4]: categories = ['A', 'B', 'C', 'D']
     values = [3, 7, 5, 4]

     plt.bar(categories, values, color='orange')
     plt.xlabel('Category')
     plt.ylabel('Values')
     plt.title('Bar Plot')
     plt.show()
```



```
[5]: data = np.random.randn(1000)

     plt.hist(data, bins=20, color='purple', edgecolor='black')
     plt.xlabel('Value')
     plt.ylabel('Frequency')
     plt.title('Histogram')
     plt.show()
```



RESULT:

Thus, the given program was written and executed successfully.